SHETH L.U.J. AND SIR M.V. COLLEGE

DEPARTMENT OF COMPUTER SCIENCE ■ ACADEMIC YEAR: 2026–2027

CampusLive: Real-Time Lecture Monitoring and Verification for Colleges

Field Project Documentation ■ S.Y. B.Sc. Computer Science (Semester III)

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Class & Semester	S.Y. B.Sc. Computer Science (Semester III)
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1. Project Title & Official Descriptions

Exact Title: CampusLive: Real-Time Lecture Monitoring and Verification for Colleges

Description: *"CampusLive is a real-time academic monitoring platform where teachers make lectures officially live, students confirm they are present with a class code, and principals and HODs see which lectures are actually being held across the college as they happen. Timetables, substitutions, reports and activity logs run on the same platform."*

Short Description: *"CampusLive is a real-time platform where teachers make lectures live, students verify them, and principals and HODs monitor what is actually being taught across the college."*

2. Guide Interaction Diary

Meeting Date	Progress / Agenda Discussed	Remarks / Guide's Feedback
06-08-2026	Initial project ideation: Identifying the need for real-time institutional visibility of ongoing lectures; project scope and objectives approval.	Recommended focusing on administrative monitoring, departmental hierarchy, and student verification.
11-08-2026	Requirement specifications: Defining the 5-tier role structure (Root -> Principal -> HOD -> Teacher -> Student) and designing the relational database schema.	Approved database entities and recommended incorporating automated conflict checks for rooms and teachers.
18-08-2026	Backend REST API design in Flask: Implementation of session authentication, PBKDF2 password hashing, and SQLite schema with Write-Ahead Logging (WAL).	Reviewed authentication security and classroom scheduling validation logic.
25-08-2026	Integration of Server-Sent Events (SSE) for push synchronization and development of the background sweeper daemon for automated lecture lifecycle tracking.	Verified real-time event distribution and background state transitions.
01-09-2026	Frontend single-page application (SPA) design: Implementing the live departures-style board, modal dialog registry, and the 4-digit student verification check-in.	Approved the student verification model and recommended keeping aggregate attendance statistics accessible to staff only.
08-09-2026	Implementation of the multi-week recurring timetable generator, HOD substitution workflow, dynamic room reallocation, and CSV reports.	Advised verifying clash resolution logic across multi-week recurring slots.
05-10-2026	End-to-end integration testing (tests/test_flow.py), field feedback review, edge-case fixes, and preparation of final documentation.	Approved the system and results for final documentation and academic submission.

3. The Project Itself & Problem Statement

CampusLive is a real-time academic activity management and operational visibility platform designed for collegiate institutions. In conventional college management, institutional leadership (Principals and HODs) lacks immediate operational visibility into daily lecture conduction without conducting physical floor rounds or manual inquiries. Furthermore, traditional paper registers consume valuable instructional time and require significant manual transcription.

CampusLive bridges this gap by transforming static timetables into an active, verifiable digital operational environment:

- **Live Institutional Visibility:** Centralized departures-style board displaying active lectures, occupied classrooms, and start punctuality.
- **Attendance and Student Verification of Lectures:** Ephemeral 4-digit class code on the teacher's screen; students confirm presence; verified at >= 2 check-ins.
- **Automated Conflict Resolution:** Real-time checks during scheduling prevent room and faculty overlapping clashes.
- **Automated Lifecycle Closure:** Background sweeper thread auto-closes forgotten live lectures after 15 min grace and updates unstarted lectures to MISSED.
- **Reports and Analytics:** Provides objective metrics: Held Rate (%), Average Start Delay (minutes), and Student Attendance (%) with CSV download.

Users and Beneficiaries

The primary beneficiary of CampusLive is the College Administration (Principal, Vice-Principals, and HODs) for centralized governance and quality assurance, alongside faculty and students:

- **Principal (College Admin):** Bird's-eye view of all academic activity, active classrooms, faculty schedules, and compliance analytics.
- **Heads of Department (HODs):** Departmental oversight, assigning substitute faculty for pending lectures, and room reallocations. Head of Department: Asst. Prof. Pradnya Kharade.
- **Teachers / Faculty:** Schedule sessions, launch live lecture, display verification PIN, track attendance, and manage weekly timetables. Core Faculty: Asst. Prof. Rohit Sahu (Project Guide) and Asst. Prof. Jyoti Chauhan (Course Coordinator).
- **Students:** View active sessions for their class, submit PIN to confirm attendance, and view timetable and attendance history. Registered Student Cohort: Aditya Rana (S105), Mohi Pathan, Suraj, Dhruv Choubey, and Akshat Halwai.
- **Root Administrator:** Provisions college instances and assigns Principal accounts.

4. Key Objectives, Modules & Technical Architecture

- **Live Campus Monitoring Board:** Real-time board grouping lectures by department with elapsed timers and late-start alerts.
- **Attendance and Student Verification of Lectures:** 4-digit code check-in; verification threshold at >= 2 check-ins; student privacy enforced.
- **Reports and Analytics:** Total, Held, Missed, and Cancelled lectures, Held Rate (%), Average Start Delay (min), and Attendance (%) with CSV export.
- **Lecture State Machine & Sweeper Daemon:** State flow SCHEDULED -> LIVE -> COMPLETED / CANCELLED / MISSED; 30-second interval daemon.
- **Weekly Timetable & Clash Resolution Engine:** Multi-week generator (1-12 weeks) skipping room and teacher schedule overlaps.
- **Academic Administration & Substitution:** HOD teacher substitution on unstarted lectures with audit tracking (covering_for) and room shifts.
- **Security, Multi-Tenant RBAC & Activity Logs:** PBKDF2 hashing, 5-tier role protection, account deactivations, and immutable event logs.

Technology Stack & Scope

Languages: Python 3.13, Modern JavaScript (ES6+), HTML5, CSS3. | **Backend:** Flask 3.x, Werkzeug Security.
Real-Time Push: Server-Sent Events (SSE) via queue.Queue, threading.Lock. | **Database:** SQLite 3 WAL Mode.
Frontend: Vanilla JavaScript Single-Page Application (SPA). | **Scope Limitations:** Remote video streaming, biometric turnstiles, student fee accounting, and native mobile store wrappers are outside functional scope.

5. Field Work, Methodology & Testing

Requirements Gathering & Field Consultation: Conducted during August 2026 at Sheth L.U.J. and Sir M.V. College:
• **Faculty and Departmental Interviews:** Interviewed 2 faculty members (1 Head of Department and 1 teaching faculty). Discussions established the operational need for an immediate link between classrooms and administrative offices and highlighted the value of automating manual paper attendance.
• **Classroom Workflow Observation:** Observed standard roll-call routines in undergraduate classrooms, noting the time consumed by manual roll calls and confirming the need for a rapid digital check-in completed in under 15 seconds.
• **Administrative Metric Definition:** Consultations with project guides and coordinators determined the essential daily tracking metrics: verifying lecture conduction, start punctuality, and generating structured attendance records.

Build and Testing Methodology: Developed using iterative agile prototyping. Automated testing with tests/test_flow.py validated 13 integration scenarios: tenant onboarding, departmental setup, conflict detection, live activation, 4-digit PIN check-in, verification threshold (>= 2 check-ins), HOD substitution, sweeper auto-completion, and CSV export. All checks passed successfully.

Future Improvements: Dynamic rotating QR codes, automated browser web push notifications, interactive visual charting (Chart.js) for Reports and Analytics, and PostgreSQL deployment profiles.

6. Evidence, Database & API Specifications

List of Figures: Fig 1: Public Landing Page | Fig 2: Student Self-Registration Modal | Fig 3: Principal Live Campus Monitoring Dashboard | Fig 4: Lecture Detail & Timeline Modal | Fig 5: Department & HOD Management | Fig 6: Faculty Directory & Live Status | Fig 7: Lecture Scheduling Modal | Fig 8: Weekly Timetable Slot Manager | Fig 9: Teacher Active Lecture Controller | Fig 10: Student Live Board & PIN Check-in | Fig 11: HOD Substitution & Room Reallocation | Fig 12: Central Activity Audit Log | Fig 13: Academic Compliance & Punctuality Report | Fig 14: User Profile & Security Settings.

Database Schema (SQLite WAL Mode): Tables: colleges (code UNIQUE), depts, classes, users, lectures, attendance (PRIMARY KEY lecture_id, student_id), timetable, logs.

Key API Endpoints: Auth (/api/login, /api/register, /api/me), Monitoring (/api/live, /api/upcoming, /api/history), Teaching & Lifecycle (/api/lectures, /api/lectures/<id>/start|end|cancel|room|substitute), Timetable (/api/timetable), Reports (/api/reports, /api/reports.csv), Real-Time Stream (/api/stream).

7. Literature Review References & Feedback Summary

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• Grinberg, M. (2018). Flask Web Development: Developing Web Applications with Python (2nd ed.). O'Reilly Media.
• Kumar, S., & Sharma, M. (2022). 'Design and Implementation of Academic Information Systems for Real-Time Institutional Monitoring.' IEEE TLT, 15(4), pp. 412–421.

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